

1. Administrative & Study Identification

Full Study Title: A Phase I/IIa Study to Investigate the Safety, Tolerability, Pharmacokinetics, and Efficacy of AZD0022 as Monotherapy and in Combination With Anti-cancer Agents in Adult Participants With Tumours Harboring a KRASG12D Mutation **Official Regulatory Title:** A Phase I/IIa, Open-label, Multi-centre Study to Assess the Safety, Tolerability, Pharmacokinetics, and Preliminary Efficacy of AZD0022 Monotherapy and in Combination With Anti-cancer Agents in Participants With Tumours Harboring a KRASG12D Mutation (ALAFOSS-01) **Short Title/Acronym:** ALAFOSS-01 **Protocol Number:** D7080C00001 **NCT Number:** NCT06599502 **Posting ID:** 1101525519562915647 **Sponsor/Company Name:** AstraZeneca **Investigational Product:** AZD0022 (oral KRASG12D-selective inhibitor) and Cetuximab **Phase of Development:** Phase I/IIa (PHASE1, PHASE2) **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine if treatment with AZD0022 as a monotherapy and in combination with other anticancer agents (such as Cetuximab) is safe and tolerable through the assessment of adverse events (AEs), serious adverse events (SAEs), dose-limiting toxicities (DLTs), and discontinuations due to toxicity. Also, to evaluate Objective Response Rate (ORR) in the efficacy expansion phase. **Secondary Objectives:** To evaluate preliminary anti-tumour activity, including Complete Response (CR) rate, Duration of Response (DoR), Disease Control Rate (DCR), and Progression-Free Survival (PFS), as well as to assess the pharmacokinetics (PK) and pharmacodynamics (PD) of the investigational regimens.

2.2 Background & Rationale To address the unmet medical need in patients with advanced solid tumors (specifically Non-Small Cell Lung Cancer [NSCLC], Colorectal Cancer [CRC], and Pancreatic Ductal Adenocarcinoma [PDAC]) harboring the historically "undruggable" KRASG12D mutation. AZD0022 is a potent oral KRASG12D-selective inhibitor that demonstrates robust pathway inhibition in preclinical models, with enhanced therapeutic responses and sustained tumor regressions observed when combined with the EGFR-targeting monoclonal antibody, cetuximab.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (First-in-human, modular, multi-centre study) **Control Type:** Single-arm/Open-label cohorts **Blinding:** Open-label **Randomization:** N/A (Sequential assignment for Dose Escalation [Part A], Dose Optimisation [Part B], and Potential Efficacy Expansion [Part C])

3.2 Planned Enrollment & Duration Target **N**: 17 participants **Number of Sites:** Multi-centre (Global) **Estimated Study Start:** 18-Oct-2024 **Estimated Primary Completion:** 20-Oct-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years or the legal age of consent in the participating jurisdiction. 2. Pathologically documented locally advanced or metastatic solid tumours (NSCLC, CRC, or PDAC) harbouring a KRASG12D mutation via tissue or liquid biopsy. 3. Participants must have progressed on, are refractory to, or are intolerant to standard of care (SoC) treatment (e.g., at least one prior line of platinum-based chemotherapy/immunotherapy for NSCLC; 2-3 prior lines for CRC). 4. Provision of a FFPE tumour sample and at least one measurable target lesion per RECIST v1.1.

4.2 Exclusion Criteria 1. Prior exposure to any direct small molecule KRAS inhibitor. 2. Symptomatic heart failure (New York Heart Association class ≥ 2) or baseline Left Ventricular Ejection Fraction (LVEF) below the institutional lower limit of normal or $< 50\%$. 3. Any evidence of clinically significant current or prior Interstitial Lung Disease (ILD) or conditions requiring IV steroids/high supplemental oxygen. 4. Any concomitant medications that are known strong inhibitors or inducers of CYP3A4/5, sensitive CYP3A4/5 substrates, or herbal preparations/medications.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: AZD0022 administered orally; escalating dosages are evaluated across sequential study modules (Module 1: Monotherapy; Module 2: Combination with Cetuximab). **Investigational Products:** * **AZD0022:** Oral tablet * **Cetuximab:** Intravenous infusion (Trade Name: Erbitux | ATC Code: L01FE01 | EMA Product Information: [EMA URL](#)) **Duration of Treatment:** From the first dose until disease progression, unacceptable toxicity, or discontinuation for any reason; approximate duration of 2 years.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications that do not interact with study pathways. **Prohibited:** Herbal preparations/medications, strong or

moderate inhibitors/inducers of CYP3A4/5, and sensitive CYP3A4/5 substrates. Receipt of any cytotoxic/non-cytotoxic drug within 21 days (or 5 half-lives) and biological therapies within 28 days prior to the first dose of study intervention is strictly forbidden.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumour KRASG12D Mutation Confirmation, LVEF/ECG Assessments
Baseline	Day 1	Enrolment, First Dose of AZD0022 ($\pm$ Cetuximab), baseline PK/PD sampling
Interim	Predefined Intervals	Safety Labs, Efficacy Assessments (Tumour Imaging via RECIST v1.1), ongoing PK/PD profiling
End of Study	Up to ~2 Years	Final Vital Signs, Exit Tumor Assessment, Safety Follow-up, Survival Status

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- 1. Safety and Tolerability (Parts A & B):** Incidence of participants with Dose-Limiting Toxicity (DLT), Adverse events (AEs), and Serious Adverse Events (SAEs) as well as changes from baseline in laboratory parameters, vital signs, and ECGs.
- 2. Discontinuation Rate (Parts A & B):** Number of patients who discontinue AZD0022 due to toxicity.
- 3. Objective Response Rate (ORR) (Part C):** Percentage of participants with a confirmed Complete Response (CR) or Partial Response (PR) evaluated according to RECIST v1.1 in the Potential Efficacy Expansion phase.

7.2 Secondary & Exploratory Endpoints

- * **Complete Response (CR) rate**
- * **Duration of Response (DoR)**
- * **Disease Control Rate (DCR)**
- * **Progression-Free Survival (PFS)** assessed radiologically per RECIST v1.1 criteria.
- * **Pharmacokinetics**, including area under the curve (AUC), maximum observed drug concentration (C_{max}), and time to reach maximum concentration (T_{max}).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via onsite visits and patient symptom monitoring. Particular attention is placed on cardiovascular toxicities (bradycardia, hypotension) and dermatological/gastrointestinal toxicities associated with Cetuximab.

Serious Adverse Events (SAEs): Reported within 24 hours to the sponsor and regulatory authorities.

Data Monitoring Committee (DMC): An internal Safety Review Committee (SRC) will evaluate dose-limiting toxicities and guide dose-escalation decisions during Parts A and B of the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: The Safety Analysis Set includes all participants who receive at least one dose of the investigational product. The Efficacy Analysis Set includes participants with measurable disease at baseline who received study intervention.

Sample Size Justification: A target $N=17$ is statistically planned for the modular Phase I/IIa design to power safety signal detection and preliminary efficacy estimates across the specified cohorts. **Handling of Missing Data:** For primary safety endpoints, missing data will generally not be imputed. For secondary survival endpoints, standard oncology censoring rules will be applied at the last known date of adequate tumor assessment.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring visits to ensure safety, accurate eCRF entry, and adherence to the FIH modular dosing protocol. **Audit Trail:** Stringent eCRF locking processes, radiological central/local reviews, and data clarification forms via standard clinical data management systems.

11. Study Identifiers & Governance

Primary ID: D7080C00001 **Registry Number:** NCT06599502 **Posting ID:** 1101525519562915647 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** ALAFOSS-01

12. Clinical Framework & Ontologies (KRASG12D Oncology Specific)

Disease Areas: Advanced Solid Tumours, Non-Small Cell Lung Cancer (NSCLC), Pancreatic Ductal Adenocarcinoma (PDAC), Colorectal Cancer (CRC) **Condition/Morphology Name:** Adenocarcinoma **Preferred UMLS Name:** KRAS G12D-positive solid tumor **Semantic Type:** Neoplastic Process **MeSH Code:** D000230 **SNOMED ID:** 1187332001 **Diagnosis Threshold:** Confirmed KRASG12D mutation via tissue or liquid biopsy **Medical Methodology:** Selective KRASG12D inhibition (AZD0022) ± EGFR inhibition (Cetuximab) **Optimization Target:** Maximal tolerated pathway suppression and tumor regression

13. Study Procedures & Methodology

Management Pathway: Targeted investigational oncology therapy **Education Protocol (HCP):** Site initiation visits and FIH

protocol-specific toxicity management (e.g., cardiac event and ILD monitoring) **Education Protocol (Patient):** Informed consent review, investigational product dosing instructions, and strict AE symptom reporting **Quality Audit Mechanism:** Safety Review Committee checkpoints and standardized tumor assessments via RECIST v1.1

14. Observation & Follow-Up Schedule

Total Duration: Approximately 2 years (from first dose to disease progression/death) **Onsite Visit Cadence:** Baseline, predefined administration intervals during treatment cycles, and End of Treatment **Remote Monitoring Frequency:** Standard safety checks and survival follow-up tracking **Checklist Review Interval:** Continuous clinical safety reviews between sequential dose-escalation cohorts